

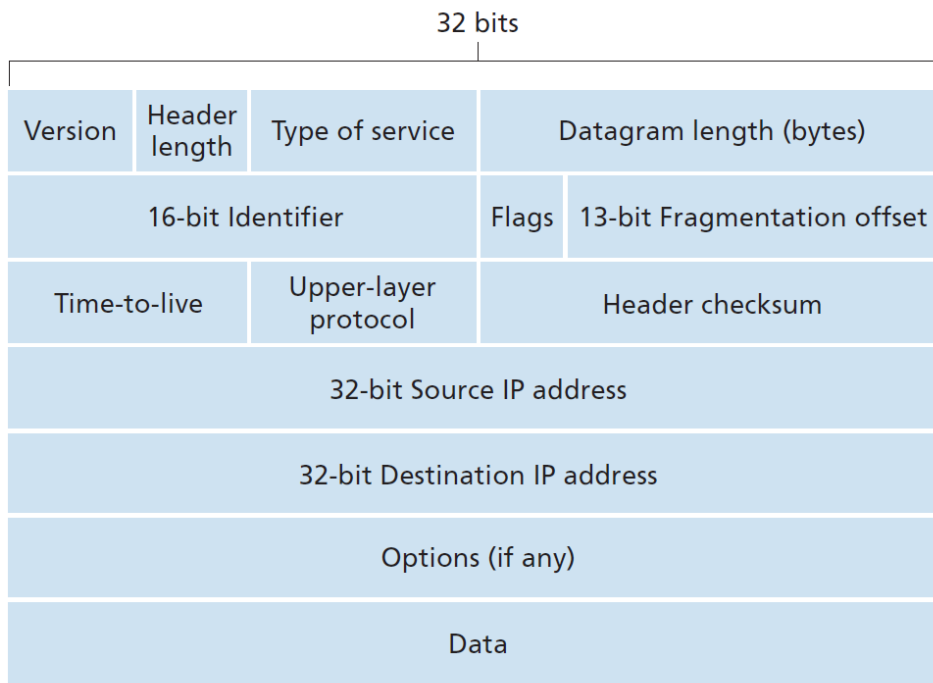
IPv4 Datagram Format

Overview

The IPv4 datagram is the fundamental packet format used throughout the Internet. Each router, switch, and networking device must understand and process this format. These notes provide a deep, clear, and modern explanation of every IPv4 header field.

1 IPv4 Datagram Structure

Below is the typical IPv4 datagram format:



1.1 Header Fields and Descriptions

1. Version Number (4 bits)

Indicates the IP protocol version. For IPv4, this value is always **4 (0100)**. Routers use this field to correctly interpret the remaining header fields.

2. Header Length (IHL, 4 bits)

Specifies the IPv4 header size in 32-bit words.

- Minimum: 20 bytes (no options)
- Maximum: 60 bytes (with options)

Routers use this to locate where the payload begins.

3. Type of Service (TOS) / DSCP

Provides Quality of Service (QoS) control. Includes:

- Priority differentiation (real-time vs normal)
- ECN (Explicit Congestion Notification)
- Traffic class identification

Used by routers for traffic shaping and congestion management.

4. Total Length (16 bits)

Indicates the entire packet size including header and data.

$$0 \leq \text{Total Size} \leq 65535 \text{ bytes}$$

Most IPv4 packets are smaller than **1500 bytes** to fit inside an Ethernet frame.

5. Identification, Flags, Fragment Offset

These fields support fragmentation.

- **Identification:** Unique per packet to reassemble fragments.
- **Flags:**
 - DF = Don't Fragment
 - MF = More Fragments
- **Fragment Offset:** Position of fragment in original packet.

IPv6 removes router-side fragmentation entirely.

6. Time-To-Live (TTL)

Prevents infinite looping.

- Decrement by 1 at every router hop.
- If TTL reaches 0, packet is dropped.
- Source receives an ICMP “Time Exceeded”.

Vital for routing stability and loop control.

7. Protocol Number

Specifies the upper-layer protocol:

- 6 = TCP
- 17 = UDP
- 1 = ICMP

This binds the network layer to the transport layer.

8. Header Checksum

Used to detect errors in the **header only**. Routers recompute the checksum at each hop because TTL changes.

9. Source and Destination IP Addresses

Source IP: Sender’s address.

Destination IP: Final receiver’s address, usually obtained via DNS.

Routers only use the **destination address** for forwarding.

10. Options Field

Optional and variable length.

Used for:

- Route recording
- Timestamping
- Security and debugging

Slows down processing; therefore removed in IPv6.

11. Data (Payload)

Carries the upper-layer segment (TCP/UDP) or ICMP data.

Typical overhead:

- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Total overhead = 40 bytes

2 Summary

Key Takeaways

- IPv4 uses a 20–60 byte header with flexible options.
- Includes support for fragmentation, QoS, error detection, and routing control.
- TTL, checksum, and protocol fields are critical for reliable operation.
- IPv6 simplifies the header by removing fragmentation and options.